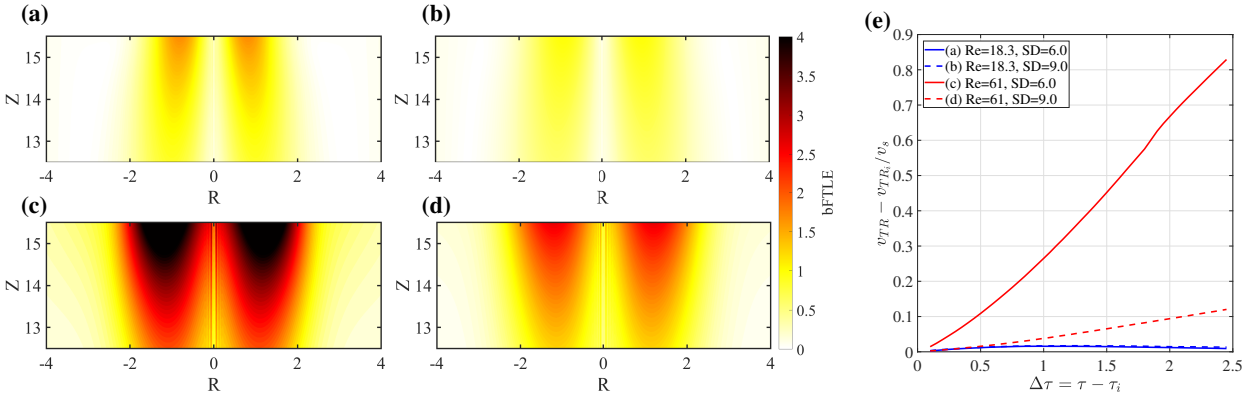

Figure 2: Backward FTLE fields with $\Delta\tau = 2.5$ for rising bubble cases C_4 ($\mathrm{Re} = 18.3$) and C_7 ($\mathrm{Re} = 61$) at different separation distances. Panels (a) and (b) correspond to case C_4 with $S = 6$ and $S = 9$, respectively, while panels (c) and (d) correspond to case C_7 with $S = 6$ and $S = 9$. Panel (e) shows the acceleration of the trailing bubble case C_4 behind the leading bubble over the same time interval $\Delta\tau = 2.5$, measured at the same vertical location where the bFTLE fields are evaluated.